

Prince Patel

+91 9979378762 | patelprince2061@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Sardar Vallabhbhai National Institute of Technology

Bachelor of Technology in Computer Science & Engineering — CGPA: 7.10

Surat, India

August 2024 – May 2028

PROJECTS

Zinthos — Music Search Engine | C++17, Rust, SQLite, FAISS

April 2026 – June 2026

- Architected a checkpointed C++17 ETL that normalized 266 GB across 3 SQLite/CSV sources into a 10-table, 145 GB catalog of 255M tracks and 348M artist links in 62.4 minutes
- Eliminated a random-I/O bottleneck by packing 256M text-to-integer ID mappings into a 6.7 GB sorted array for binary-search joins, remapping 254.8M audio rows at 294K rows/s within 15 GB RAM
- Built a resumable, out-of-core PyTorch/FAISS pipeline using keyset pagination, memory-mapped files, and atomic batch checkpoints; encoded 254.8M tracks in 21.7 minutes at 220K rows/s
- Selected a 9.17 GB fp16 IVF index with 94.3% recall@10 over a 29.5%-recall PQ design, then memory-mapped it for CPU serving with two-stage retrieval and feature-based reranking
- Shipped 4 interactive workflows through FastAPI and a Rust Ratatui client, using background jobs and channels to keep the 100 ms UI loop responsive while serving warm queries in 0.4–60 ms

Descriptive CCTV Search | Python, PostgreSQL, FastAPI, Next.js

July 2026 – Present

- Architected a surveillance-search stack indexing 41.5K tracklets and 74K multi-view crops from 14 scene-camera streams as 1152-D SigLIP vectors in a 1.7 GB PostgreSQL/pgvector database
- Accelerated fixed-camera ingestion 71% ($0.56\times$ to $0.96\times$ real time) by disabling unnecessary BoT-SORT optical-flow compensation and capping CPU pools, cutting 12-core load average from 25 to 7.7
- Streamed 14.9K live tracklets through a 4-stage bounded pipeline with 8-frame/256-tracklet backpressure, per-tracklet commits, and 2-worker asynchronous CPU OCR off the GPU path
- Benchmarked 4 re-identification encoders on 737 cross-camera queries, retaining FastReID at 47.1% rank-1/63.2% rank-5 after DINOv2 reached 10.9% and VeRi-Wild 38.0%
- Shipped 4 search modes and 5 case APIs with immutable audit trails; recorded 61 searches and 7 append-only export receipts for SHA-256/Ed25519-signed packages up to 623 MB

TECHNICAL SKILLS

Languages: C/C++, Python, Rust, TypeScript, SQL, Bash

Systems & Tooling: Linux, Git, Docker, CMake, Nix

Data & Web: SQLite, PostgreSQL, pgvector, FastAPI, Next.js

Machine Learning: PyTorch, FAISS, RAPIDS cuML, LightGBM, OpenCV, Ultralytics/YOLO, Transformers

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, Database Management Systems, Computer Organization & Architecture, Object-Oriented Programming, Discrete Mathematics, Probability & Statistics